

E's gravity.

E & M.

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G_N T_{\mu\nu}$$

$$h_{\mu\nu}^{\text{new}} = h_{\mu\nu}^{\text{old}} - \xi_{\mu\nu} - \xi_{\nu\mu}$$

$$X_{\text{new}}^{\mu} = X_{\text{old}}^{\mu} - \xi^{\mu}$$

$$\text{Choose } \xi^{\mu} : h_{\mu}^{\nu}{}_{;\nu} - \frac{1}{2}h_{;\mu} = 0$$

Choose a coordinate to make our lines easy
 & some things cancel out

Simplifying the.

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \text{ eqn}$$

$$= \frac{1}{2} (h^{\alpha}{}_{\mu\nu;\alpha} + h^{\alpha}{}_{\nu\mu;\alpha} - \square h_{\mu\nu} - h_{;\mu\nu})$$

$$+ \frac{1}{2}\eta_{\mu\nu}(h^{\alpha\beta}{}_{;\alpha\beta} - \square h)$$

$$h^{\alpha}{}_{\mu\nu;\alpha} = \frac{1}{2}h_{;\mu\nu}$$

$$h^{\alpha\beta}{}_{;\alpha\beta} = h^{\alpha\beta}{}_{;\beta\alpha} = \frac{1}{2}h^{\alpha}{}_{;\alpha} = \frac{1}{2}\square h$$

simplifies to

$$= -\frac{1}{2}(\square h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\square h)$$

N.R. limit.

$$\square h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\square h = -16\pi G_N T_{\mu\nu}$$

$$|T_{ij}| \ll |T_{0i}| \ll |T_{00}|$$

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

assume time deriv $\ll$ spatial deriv

$$\square \approx \nabla^2$$

$$\partial_{\nu} F^{\mu\nu} = j^{\mu}$$

$$F^{\mu\nu} = \partial^{\mu} A^{\nu} - \partial^{\nu} A^{\mu}$$

$$A_{\text{new}}^{\mu} = A_{\text{old}}^{\mu} - \partial^{\mu} \chi$$

arbitrary

$$F_{\text{new}}^{\mu\nu} = F_{\text{old}}^{\mu\nu} + \partial^{\mu} \partial^{\nu} \chi - \partial^{\nu} \partial^{\mu} \chi$$

$$\partial^{\mu} \partial_{\nu} A^{\nu} - \square A^{\mu} = j^{\mu}$$

$$0 = \partial_{\mu} A_{\text{new}}^{\mu} = \partial_{\mu} A_{\text{old}}^{\mu} - \square \chi$$

simplify

$$\square A^{\mu} = -j^{\mu}$$

$$\begin{pmatrix} \phi \\ \vec{A} \end{pmatrix}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_{electric}}{\epsilon_0}$$

$$\Leftrightarrow \nabla^2 \phi_{ec} = -\frac{\rho_{ec}}{\epsilon_0}$$

$$F_{coulomb} = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$$

$$F_{newton} = -G_N \frac{Mm}{r^2}$$

$$\nabla^2 \phi_{grav} = 4\pi G_N \rho_{mass}$$

$$\square (h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h) = 16\pi G_N T_{\mu\nu}$$

$$\eta^{\mu\nu} (\dots) = \eta^{\mu\nu} (\text{trace of } T)$$

$$\square (h - \frac{4}{2} h) = -16\pi G_N (\eta^{\mu\nu} T_{\mu\nu})$$

$$\nabla^2 h \approx \square h = -16\pi G_N \rho_{mass}$$

$$h = -4\phi_{grav}$$

0-0 component: $\square (h_{00} + -\frac{4}{2} \phi_{grav}) = -16\pi G_N \rho$

$$h_{00} = -2\phi_{grav}$$

1-1 component:

$$\square (h_{11} - \frac{(-4)}{2} \phi_{grav}) = 0$$

$$h_{11} = -2\phi_{grav}$$

$$\eta + h = \begin{pmatrix} -(1+2\phi) & & & \\ & 1-2\phi & & \\ & & 1-2\phi & \\ & & & 1-2\phi \end{pmatrix}$$

$$\phi = \frac{G_N M}{r}$$

physicist idea!

geodesic

minimizes

this is the correct newtonian

limit $\Rightarrow$ that explains why

the coefficient has to be 8/3

proper distance

In the vacuum:

$T_{\mu\nu} = 0, h = 0$

$\square(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h) = -16\pi G_N T_{\mu\nu} = 0$

$\Downarrow$

$\square h_{\mu\nu} = 0$

More analogies:

gravity

$\square h_{\mu\nu} = 0$

$h_{\mu\nu} = \epsilon_{\mu\nu} e^{ikx} + c.c.$

$\square h_{\mu\nu} = 0 \Rightarrow k^2 = 0$

by a tensor:
direction indicated

2 physical polarizations

E & M.

$\square A_\mu = 0$

$A_\mu = \epsilon_\mu e^{ikx}$

$kx \equiv k_\mu x^\mu, k^\mu = \begin{pmatrix} k^0 \\ \vec{k} \end{pmatrix}$

$kx = k^0 t + \vec{k} \cdot \vec{x} = -k^0(t - \hat{k} \cdot \vec{x})$

$A_\mu = \epsilon_\mu e^{-ikx}$

$\square A_\mu = \epsilon_\mu (k^0)^2 k^x e^{ikx} + c.c.$

$\Rightarrow k^2 = 0$

moves with speed of light.

like $\epsilon_\mu = (0, 0, 1, 0)$

which way the electric field points

polarization vector

2 polarizations of E & M.